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How does biodiversity risk affect digital transformation? Evidence from China's manufacturing enterprises

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ABSTRACT

This paper examines the impact of biodiversity risk on digital transformation by using the data of Chinese A-share listed companies in manufacturing industry from 2011 to 2023. The findings indicate that biodiversity risk significantly promotes digital transformation. Furthermore, we find corporate ESG performance and R&D investment play mediating role between biodiversity risk and digital transformation. Our findings offer novel insights into how biodiversity risk enhances digital transformation of enterprises.

1. Introduction

Biodiversity risk has emerged as a critical and underexplored threat to global economic stability, with annual losses estimated at \$20 trillion per year (Giglio et al., 2023). While physical risks such as habitat degradation are pressing, transition risks—including regulatory, reputational, and market shifts driven by biodiversity loss—are increasingly reshaping corporate asset valuations and strategic priorities (Pi et al., 2025). In response, regulatory frameworks are evolving. Notably, in 2024, China issued the "Self--Regulatory Guidelines for Listed Companies——Sustainable Development Report (Trial)", which for the first time include the topic of biodiversity protection in order to guide listed companies in understanding and reporting on sustainable development performance.

Manufacturing enterprises, due to their resource intensity, supply chain complexity, and environmental footprint, are especially vulnerable to biodiversity risks (Wagner, 2023). On one hand, increasingly stringent environmental regulations and carbon emission constraints are elevating both operational costs and compliance burdens (Li et al., 2025). On the other hand, rising societal and consumer expectations are intensifying scrutiny of firm's ecological footprints (Kopnina et al., 2024), posing risks to brand value and long-term competitiveness. In response to these mounting pressures, firms are compelled to seek innovative strategies for risk mitigation and adaptation. In this context, digital transformation plays an important role in tackling biodiversity risk (Flak, 2020). Technologies such as big data analytics can help optimize production efficiency, while IoT-based monitoring and blockchain-enabled supply chains enable firms to trace, disclose, and mitigate biodiversity-related risks (Truong, 2022). Therefore, understanding how manufacturing enterprises adapt to biodiversity risk and how biodiversity risk affects the digital transformation of enterprises are important and timely research questions.

To investigate the impact of biodiversity risk on the digital transformation of enterprises, we analyze the data of Chinese listed companies in manufacturing industry from 2011 to 2023. Regression results indicate that biodiversity risk promotes digital transformation of enterprises. After a series of robustness tests, the results remain robust. Finally, the mechanism analysis reveals that corporate Environmental, Social, and Governance (ESG) performance and research and development (R&D) investment play

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mediating effect between the biodiversity risk and digital transformation of enterprises.

The contributions of this paper are as follows:

Firstly, we provide empirical evidence that biodiversity risk positively enhances digital transformation of enterprises, thereby enriching digitalization influencing factors by incorporating a novel environmental dimension of biodiversity risk. This is important, because prior studies focus more on climate factors and enterprise digital transformation (eg., Chen and Zhang, 2025; Mo and Liu, 2023). **Secondly,** we identify ESG performance and R&D investment as key mediating channels, showing how firms convert ecological risks into strategic digitalization. This highlights how internal governance actions can foster corporate resilience and proactivity in the face of environmental challenges. **Finally,** we reveal heterogeneous effects of biodiversity risk on enterprises digital transformation across geographic regions and levels of environmental policy intensity, offering valuable insights for policymakers to design targeted incentive measures that align with regional characteristics.

2. Literature review and hypothesis development

2.1. The impact of biodiversity risk on digital transformation of enterprises

Biodiversity risk refers to the potential threats that the loss of biodiversity and the degradation of ecosystems pose to various stakeholders (Dempsey, 2013). Manufacturing enterprises, given their significant environmental footprint, are particularly vulnerable to biodiversity risk (Wagner, 2023). This risk manifests as multifaceted pressure: from regulators imposing stricter environmental standards (Li et al., 2025), and from consumers or investors demanding greater ecological responsibility (Kopnina et al., 2024). From the perspective of **stakeholder theory**, firms facing heightened biodiversity risk are compelled to signal their commitment to sustainability to maintain legitimacy and social license to operate. In response to this complex and uncertain environment, digital transformation emerges as a critical adaptive strategy (Mo and Liu, 2023), and serves as a visible and substantive way to demonstrate this commitment to key stakeholders. Moreover, under the lens of **dynamic capabilities theory**, biodiversity risk introduces profound environmental uncertainty, which compels manufacturing enterprises to enhance their technological capabilities and operational agility. Technologies such as big data analytics and blockchain-enabled supply chains can optimize resource use and enable real-time environmental monitoring, helping firms address both operational and regulatory challenges. Thus, digital transformation equips manufacturing enterprises with the capabilities needed to navigate the uncertainties associated with biodiversity risk. Therefore, we propose the first hypothesis:

Hypothesis 1. Biodiversity risk positively enhances digital transformation of enterprises.

2.2. The mediating effect of ESG performance

Firms exposed to biodiversity risk often face operational disruptions stemming from supply chain instability, resource scarcity and reputational damage (Lähtinen et al., 2016). In more severe cases, such risks can increase the likelihood of stock price volatility and bankruptcy (Adamolekun, 2024). Moreover, growing stakeholder awareness of environmental sustainability. These pressures incentivize firms to adopt proactive strategies to mitigate ecological risks. Nowadays, investors are increasingly preferring to pay a premium for investments that support ecology conservation (Pi et al., 2025). Notably, strong ESG performance at the firm level has been shown to effectively improve corporate reputation, enhance stakeholders trust, and enhance firm operational stability (Chen and Xie, 2022; Wong et al., 2021).

From the perspective of **stakeholder theory**, in response to intensifying biodiversity risk, firms are incentivized to improve their ESG performance to meet regulatory demands and stakeholder expectations (Pi et al., 2025). In addition, achieving superior ESG ratings, particularly on the environmental dimension, requires robust data collection, monitoring, and transparent reporting (Zhao and Cai, 2023). This, in turn, drives the adoption of digital technologies (e.g., AI-powered analytics, blockchain-based supply chains) necessary for these tasks (Hao et al., 2025; Yang et al., 2025). We therefore hypothesize:

Hypothesis 2. Biodiversity risk can enhance ESG performance, thereby promoting digital transformation of enterprises.

2.3. The mediating effect of R&D investment

Strategic growth option theory aligns with the proverb, "The early bird catches the worm." This theory posits that firms are more inclined to invest aggressively during times of high uncertainty (Boah and Ujah, 2024). Drawing on strategic growth option theory, high uncertainty, such as that created by biodiversity risk, can spur proactive investment. In the competitive Industry 4.0 era, to preempt competitors and secure future market position, manufacturing enterprises are more incentivized to increase R&D spending to develop innovative and sustainable technologies (Sarbu, 2022). Because delaying investment may thus allow competitors to seize valuable opportunities. Enhanced R&D activities not only strengthen a firm's technological foundation, but also build digital capabilities (Yan et al., 2023). Specifically, R&D projects aimed at risk identification and sustainable innovation often inherently require digital tools for simulation, data analytics, and process automation (Guan et al., 2022; Zhuo and Chen, 2023), thus creating a direct bridge from research investment to digitalization infrastructure. Thus, R&D investment acts as another critical pathway linking risk to transformation. This leads to the third hypothesis:

Hypothesis 3. Biodiversity risk can stimulate R&D investment, thereby promoting digital transformation of enterprises.

2.4. Theoretical framework

The theoretical framework of the empirical study is represented in Fig. 1.

3. Research design

3.1. Methodology

To analyze the impact of biodiversity risk (BR) on enterprises digital transformation (DIGITAL), fixed-effect model is used, and regression equation is shown in Eq. (1):

$$\mathrm{DIGITAL}_{\mathrm{i,t}} = \alpha + \beta \mathrm{BR}_{\mathrm{i},t-1} + \sum \gamma \cdot \mathrm{Controls} + \mathrm{year}_{\mathrm{t}} + \mathrm{firm}_{\mathrm{i}} + \varepsilon_{\mathrm{i,t}} \tag{1}$$

To further investigate the mediating effect, we construct Eq. (2) and Eq. (3):

$$\mathrm{Mediator}_{\mathrm{i,t}} = \alpha + \beta \mathrm{BR}_{\mathrm{i},t-1} + \sum \gamma \cdot \mathrm{Controls} + \mathrm{year}_{\mathrm{t}} + \mathrm{firm}_{\mathrm{i}} + \varepsilon_{\mathrm{i,t}} \tag{2}$$

$$\mathrm{DIGITAL}_{\mathrm{i,t}} = \alpha + \beta \mathrm{BR}_{\mathrm{i},t-1} + \delta \mathrm{Mediator}_{\mathrm{i,t}} + \sum \gamma \cdot \mathrm{Controls} + \mathrm{year}_{\mathrm{t}} + \mathrm{firm}_{\mathrm{i}} + \varepsilon_{\mathrm{i,t}} \tag{3}$$

Where Mediator represents the two mediating variables: corporate ESG performance (ESG) and R&D investment (R&D).

3.2. Variable definition

3.2.1. Dependent variable: digital transformation of enterprises

Following existing literature (Yu, 2024; Zhen et al., 2023), we use the enterprise digital transformation index (DIGITAL) as the dependent variable.[1] This index is obtained from CSMAR database. A higher value of this index reflects a greater level of digital transformation within the firm.

3.2.2. Independent variable: biodiversity risk

This paper utilizes the biodiversity risk (BR) as the independent variable. BR equals 1 if the frequency of biodiversity-related terms[2] in firm's annual report is more than two, and 0 otherwise (He et al., 2024).

As shown in Fig. 2, biodiversity risk (BR) has increased steadily since 2011. This upward trend indicates that China's listed manufacturing enterprises have experienced growing exposure to biodiversity-related pressures over the sample period.

However, biodiversity risk is not distributed evenly across the manufacturing industry. As illustrated in Fig. 3, both the mean biodiversity risk (BR) and the mean enterprises digital transformation index (DIGITAL) vary significantly among different sub-categories of the manufacturing sector.[3]

3.2.3. Control variables

Following established literature, we select 10 control variables that may influence digital transformation. Detailed definition of variables is presented in appendix Table A.3.

3.3. Data sources

We select Chinese A-share listed companies in manufacturing industry in Shanghai and Shenzhen stock markets as research sample. Digital transformation index and corporate financial data are obtained from the CSMAR database. Biodiversity risk index (BR) is obtained from an open database in existing research (He et al., 2024). Corporate ESG performance data is obtained from Sino-Securities Index ESG ratings database. Samples are preprocessed in following criteria: (1) excluding ST, *ST, PT firms, and firms in the delisting consolidation period; (2) excluding firms in financial industry; (3) excluding firms with missing data; (4) applying a 1 % winsorization to numerical variables to mitigate the impact of outliers. Finally, 16,770 observations from 2011 to 2023 are obtained. Variables descriptive statistics is presented in appendix Table A.4.

[1] The DIGITAL index is constructed based on a multi-dimensional framework comprising six key indicators across two levels. At the firm level, these include Strategic Leadership, Technology-Driven Development, Organizational Empowerment, Digital Achievements, and Digital Applications; at the macro level, it incorporates Environmental Support. Compared to existing approach (e.g., Wu et al., 2021), which rely on keyword frequency in the MD&A section to measure digital transformation, our DIGITAL index offers broader coverage and richer content, enabling a more effective and comprehensive measurement of enterprises' digitalization levels. And this index is widely used in studies of Chinese listed firms. **Due to space constraints, we refer readers to Appendix Table A.1 for detailed information on the components of DIGITAL.**

[2] Biodiversity-related words are shown in appendix Table A.2.

[3] According to the 2012 industry classification guidelines issued by the China Securities Regulatory Commission (CSRC), the manufacturing sector (Code C) is divided into C13-C43 (subcategories). Details of C13-C43 are shown in Figure 4.

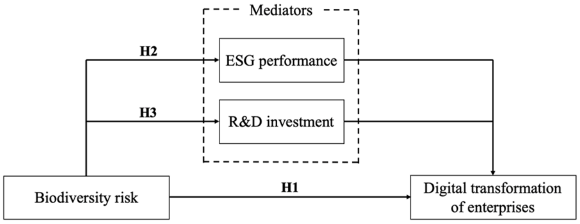

Fig. 1. Theoretical framework.

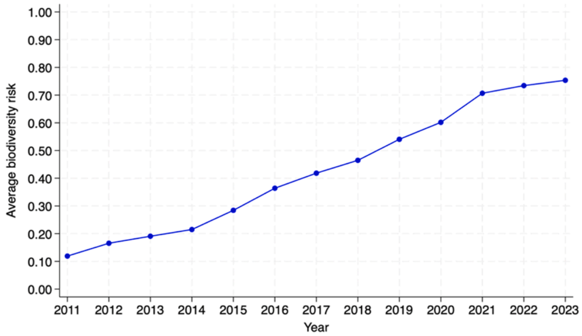

Fig. 2. Time plot for firm’s average BR.

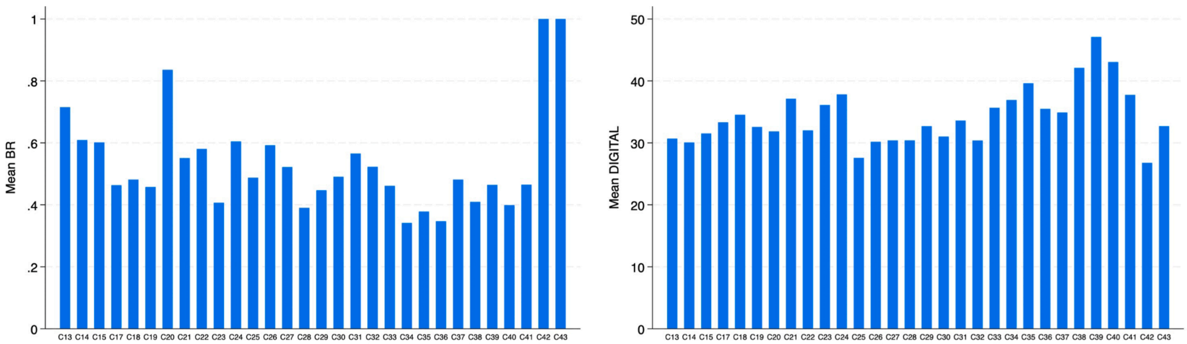

Fig. 3. Mean BR and DIGITAL by subcategories of manufacturing sector.

4. Empirical analysis

4.1. Baseline regression

As shown in Table 1, the coefficient for biodiversity risk (BR) is positive and significant at the 1 % level in all specifications. This suggests that exposure to biodiversity risk promotes digital transformation of enterprises. Specifically, in column (2), this coefficient implies that, ceteris paribus, firms exposed to biodiversity risk (BR=1) have a digital transformation index that is, on average, 0.642 points higher than that of firms not exposed to such risk. This provides initial evidence that exposure to biodiversity risk significantly promotes corporate digital transformation.

4.2. Robustness test

4.2.1. Replacing independent variable

We employ alternative measures for our independent variable. We replace the primary biodiversity risk measure (BR) with biodiversity concern (BC). The result, shown in Column (1) of Table A.5, indicates that the coefficient for BC is positive and significant.

Additionally, to test the sensitivity of our primary measure (BR) to its threshold, we create two alternative dummy variables: BR_one (if keyword frequency is > 1) and BR_three (if keyword frequency is > 3), to run regression again. As presented in Columns (2) and (3) of Table A.5, the coefficients for both BR_one and BR_three remain positive and highly significant. These results collectively

C13: Agricultural and By-Product Food Processing
C14: Food Manufacturing
C15: Liquor, Beverages, and Refined Tea Manufacturing
C16: Tobacco Products
C17: Textile Industry
C18: Garment and Apparel Manufacturing
C19: Leather, Fur, Feather, and Related Products and Footwear Manufacturing
C20: Wood Processing and Products of Wood, Bamboo, Rattan, Palm, and Straw
C21: Furniture Manufacturing
C22: Paper and Paper Products
C23: Printing and Reproduction of Recording Media
C24: Stationery, Artistic, Sporting, and Entertainment Goods Manufacturing
C25: Petroleum Processing, Coking, and Nuclear Fuel Processing
C26: Chemical Raw Materials and Chemical Products Manufacturing
C27: Pharmaceutical Manufacturing
C28: Chemical Fiber Manufacturing
C29: Rubber and Plastic Products
C30: Non-Metallic Mineral Products
C31: Ferrous Metal Smelting and Rolling
C32: Non-Ferrous Metal Smelting and Rolling
C33: Metal Products
C34: General Equipment Manufacturing
C35: Specialized Equipment Manufacturing
C36: Automobile Manufacturing
C37: Railway, Shipbuilding, Aerospace, and Other Transport Equipment Manufacturing
C38: Electrical Machinery and Equipment Manufacturing
C39: Computers, Communication, and Other Electronic Equipment Manufacturing
C40: Instrumentation and Cultural, Educational, and Office Equipment Manufacturing
C41: Other Manufacturing
C42: Recycling and Utilization of Waste Resources
C43: Repair Services for Metal Products, Machinery, and Equipment

Fig. 4. Subcategories of manufacturing sector.

Table 1
Results of baseline regression.

	(1) DIGITAL	(2) DIGITAL
BR	0.695***	0.642***
	(5.42)	(5.07)
Constant	30.549***	35.583*
	(152.84)	(1.86)
Controls	NO	YES
N	14,842	14,842
Year FE	YES	YES
Firm FE	YES	YES
R^2	0.326	0.332

Note: ***, **, and * denote significance at 1 %, 5 %, and 10 % levels.
Numbers in parentheses are robust t-value.

affirm the robustness of our main finding.

4.2.2. *Changing the measurement approach of dependent variable*

We use an alternative measure for the dependent variable, denoted as DIGITAL_replace.[4] The regression result in column (4) of Table A.5 shows that the coefficient for biodiversity risk (BR) remains positive and statistically significant, which is consistent with our baseline finding in Table 1.

4.2.3. *Excluding the samples during COVID-19 periods*

To mitigate the potential confounding effects of the COVID-19 pandemic, we exclude samples from 2020–2022 and re-run the regression. The result, presented in Column (5) of Table A.5, shows that the coefficient of BR remains significantly positive, confirming that our findings are not driven by the unique economic conditions of the pandemic period.

4.2.4. *2SLS regression*

To address potential endogeneity concerns, we employ a two-stage least squares (2SLS) regression using an instrumental variable (IV) approach. We adopt *one lag period of average biodiversity risk in the same subcategories of manufacturing sector and city* (IV-BR) as an instrumental variable. The results of 2SLS regression also indicate empirical results of this paper are robust.[5]

4.2.5. *DID regression*

Regression model. We analyze the impact of the external shock (Announcement of China's Priority Area for Biodiversity Conservation), a policy announced by China's Ministry of Ecology and Environment, by using difference-in-difference (DID) approach to mitigate potential endogeneity issues. Since this policy was announced on December 31, 2015, its substantive impact is expected to begin in the following year. Therefore, we select 2016 as the base year for the policy implementation to examine the impact of external shock. Post is assigned a value of 0 for years prior to 2016, and 1 after 2016. Next, research sample is divided into two groups: firms with a cumulative BR of less than 1 before 2016 are treated as control group, and the remaining firms are assigned to the treatment group.[6] The DID regression model is specified as follows:

$$\mathrm{DIGITAL}_{\mathrm{i,t}} = \alpha + \beta \mathrm{Treat}_{\mathrm{i,t}} * \mathrm{post}_{\mathrm{i,t}} + \delta \mathrm{Treat}_{\mathrm{i,t}} + \tau \mathrm{post}_{\mathrm{i,t}} + \sum \gamma \cdot \mathrm{Controls} + \mathrm{year}_{\mathrm{t}} + \mathrm{firm}_{\mathrm{i}} + \varepsilon_{\mathrm{i,t}} \tag{4}$$

where Treat is assigned 1 for firms in the treatment group, and 0 otherwise.

Parallel trend test. Before doing DID regression, we perform a parallel trend test, the results of which are presented in Table A.6 and are consistent with our expectations.[7]

Analysis of regression result. The DID regression result is shown in appendix Table A.7. The coefficient of the interaction term, Treat*post, is 0.547 ($t = 1.59$). This finding suggests that the treatment group (firms with higher prior biodiversity risk exposure) increased their digital transformation levels more significantly after the policy implementation. This quasi-experimental evidence further corroborates our main conclusion that biodiversity risk acts as a catalyst for digital transformation of enterprises.

5. Further analyses

5.1. *Mechanism test*

5.1.1. *Corporate ESG performance*

Columns (1) and (2) of Table 2 show that biodiversity risk significantly improves corporate ESG performance, which in turn positively influences digital transformation. The mediation effect is statistically significant, suggesting that ESG performance serves as an important conduit through which biodiversity risk drives manufacturing enterprises' digital upgrading. This finding supports Hypothesis2 and aligns with recent research indicating that improved ESG performance compel firms to accelerate their digital

[4] We utilize the DIGITAL_replace metric, a word frequency methodology developed by Wu et al. (2021), to assess the extent of digital transformation.

[5] A valid instrument must satisfy the exogeneity and relevance conditions. Firstly, the instrument variable (IV-BR) satisfies the exogeneity condition. Because peer risk disclosure influences a firm's own risk perception via peer effects, but should not directly determine its internal digital transformation strategy. Secondly, the relevance condition, which requires a strong correlation between the instrument variable and the endogenous variable, is supported by the first-stage results in column (6) of Table A.5 in appendix. The coefficient on IV-BR is highly significant (β=0.295, t = 22.58), and the Cragg–Donald Wald F statistic of 804.848 is well above the conventional threshold of 10, indicating that our chosen instrument variable is strong and not subject to the weak instrument problem. The second-stage results are presented in column (7) of Table A.5. The coefficient for BR is 2.434 and remains significant at the 1% level. **This result confirms that our main finding is robust.**

[6] A cumulative BR of 1 represents a threshold indicating that a firm has been consistently exposed to or concerned about biodiversity risk even before the policy shock, making them a suitable treatment group.

[7] The validity of the DID model relies on the parallel trend assumption, which requires that the treatment and control groups shared similar trends before the policy shock. The results in Table A.6 support this assumption: the coefficients for the pre-treatment periods (pre_3 and pre_2) are statistically insignificant, indicating no pre-existing differential trend between the groups. In contrast, the coefficients become positive and significant in the post-treatment periods, starting from the year of the shock (current).

Table 2
Mechanism tests.

	(1) ESG	(2) DIGITAL	(3) R&D	(4) DIGITAL
BR	0.043*	0.635***	0.059***	0.565***
	(1.91)	(5.03)	(3.32)	(4.41)
ESG		0.150***		
		(2.60)		
R&D				0.712***
				(5.77)
Constant	8.398***	34.324*	12.262***	25.301
	(4.41)	(1.79)	(3.47)	(1.20)
Controls	YES	YES	YES	YES
N	14,842	14,842	14,277	14,277
Year FE	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES
R^2	0.034	0.333	0.486	0.340

transformation processes.

5.1.2. R&D investment

Columns (3) and (4) in Table 2 indicate that biodiversity risk also promotes digital transformation through increased R&D investment. The mediation effect is both positive and statistically significant. This suggests that firms facing higher biodiversity risk are more inclined to invest in R&D to develop adaptive, sustainable digital solutions, enabling them to better monitor and mitigate biodiversity impacts. This result provides empirical support for Hypothesis 3.

5.2. Heterogeneity test

We conducted heterogeneity tests based on geographic location[8] and the environmental protection intensity.[9]

First, the results of the heterogeneity analysis are presented in Table 3. As shown in columns (1) and (2), the coefficient of BR in the western sample is significantly higher than that in the eastern sample. This difference can be attributed to several factors: The eastern region has a more developed economy and an industrial structure inclined towards technology-intensive manufacturing, resulting in stronger digital foundations for its manufacturing companies. Consequently, the impact of BR on DIGITAL is relatively smaller in the eastern sample. In contrast, the western region's economy is less developed, with a greater proportion of traditional manufacturing industries. These manufacturing companies face greater pressure for upgrading and transformation, leading to a more pronounced impact of BR on their digital transformation efforts.

Second, results in columns (3) to (4) show that, the coefficient of BR in the sample of the key environmental protection cites, is significantly positive and higher than that in the sample of the non-key environmental protection cites. This is driven by stringent regulations and heightened environmental awareness, which encourage manufacturing companies to adopt advanced digital technologies to address sustainability challenges. These cities often provide better access to resources, expertise, and innovation ecosystems that support digital transformations.

6. Conclusion

This study find that biodiversity risk enhances digital transformation of enterprises. The robustness test further supports the reliability of the baseline regression results. Furthermore, we confirm corporate ESG performance and R&D investment play mediating role between biodiversity risk and digital transformation of enterprises. Additionally, heterogeneity analysis reveals that this positive impact is more pronounced for firms located in China's western regions and in cities with stricter environmental regulations.

Based on these findings, we offer several practical implications for policymakers and corporate managers. For policymakers: Firstly, promote R&D-driven ecological digitalization. Given that R&D investment is a key channel through which biodiversity risk drives digital transformation, we recommend that policymakers design targeted tax credits or subsidies for corporate R&D projects that explicitly fuse digital technologies with biodiversity goals. Such incentives could be prioritized for manufacturing subcategories identified with both high biodiversity risk and lower initial digitalization levels (e.g., Wood Processing, C20; Paper and Paper Products,

[8] In terms of geographical location, we divided the sample into two categories: firms located in the eastern side of "Hu Huanyong line" and westnern side of "Hu Huanyong line". Samples are categorized according to their positions relative to the "Hu Huanyong Line", a significant demographic and geographic dividing line in China that stretches diagonally from *Heihe* (northeast city of China) to *Tengchong* (southwest city of China). This line separates China into two distinct zones: the densely populated and economically developed eastern side, and the sparsely populated, natural resource-rich western side.

[9] According to the "National Environmental Protection 11th Five-Year Plan" issued by the State Council of China in 2007, the samples are divided into two groups: firms located in key environmental protection cities and firms located in non-key environmental protection cities.

Table 3
Results of heterogeneity analysis.

	(1) Firms located in the eastern side of "Hu Huanyong" line DIGITAL	(2) Firms located in the western side of "Hu Huanyong" line DIGITAL	(3) Firms located in the key environmental protection cities DIGITAL	(4) Firms located in the non-key environmental protection cities DIGITAL
BR	0.625***	2.246***	0.672***	0.624*
	(4.51)	(3.06)	(4.57)	(1.76)
Controls	YES	YES	YES	YES
Constant	41.037*	−98.845	45.240*	−28.678
	(1.84)	(−0.84)	(1.90)	(−0.58)
N	12,455	394	11,065	1833
Year FE	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES
R^2	0.330	0.439	0.339	0.292

C22), maximizing policy impact. And this could include incentivizing the development of AI for species monitoring, IoT for pollution control, or blockchain for enhancing green supply chain transparency. Secondly, leverage regional heterogeneity for pilot Programs. The finding that firms in key environmental protection cities are more responsive suggests these areas are ideal pilot zones for launching advanced "ecological-digital" policies. Furthermore, policies should be tailored to regional economic differences, potentially offering greater support for foundational digital infrastructure in the less-developed western regions to help them overcome initial adoption hurdles.

For corporate managers: Firms are encouraged to view biodiversity risk not as a mere compliance burden but as a strategic catalyst for innovation and competitive advantage. Firstly, integrate digital tools with ESG strategy. The mediating role of ESG performance indicates that digital transformation and sustainability management should be deeply intertwined. Managers should move beyond siloed functions and embed digital tools directly into their ESG processes. For example, leveraging digital platforms for transparent sustainability reporting or using sensor technology for real-time environmental impact monitoring can simultaneously improve ESG ratings, mitigate risks, and enhance operational resilience.

Future research is encouraged to explore more mediating channels, such as green innovation capability or executive environmental awareness. Additionally, longitudinal studies could examine how evolving biodiversity regulations shape long-term digitalization trajectories across industries.

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CRediT authorship contribution statement

Gang Wang: Writing – review & editing, Writing – original draft, Methodology, Investigation, Data curation, Conceptualization.

Declaration of competing interest

There are no competing interests to declare.

Supplementary materials

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Data availability

The authors do not have permission to share data.

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